CSE 318 — Online 2

Section A1

In your Randomized heuristic, each vertex is placed into X or Y with a fair coin flip (probability $\frac{1}{2}$ each). The coin flip completely ignores the graph structure.

Your task: Replace the fair coin flip with a **smarter** coin that looks at the current partial cut. Use a **temperature** parameter $\tau > 0$ to control how much the coin cares about the graph:

1. Process the vertices in a random order (generate a random permutation of V).
2. For the current vertex v, compute:
   - $\sigma_X(v) = \sum_{u \in X} w_{vu}$ (sum of weights to vertices already in X)
   - $\sigma_Y(v) = \sum_{u \in Y} w_{vu}$ (sum of weights to vertices already in Y)
3. Place v into X with probability:

   $$p_X = \frac{1}{1 + e^{(\sigma_X(v) - \sigma_Y(v))\,/\,\tau}}$$

   Otherwise place v into Y.
4. Note: placing v in X earns $\sigma_Y(v)$ towards the cut, and placing v in Y earns $\sigma_X(v)$. When $\sigma_Y(v) > \sigma_X(v)$, the formula makes $p_X > 0.5$, i.e. it prefers the better side.

The two extreme cases:

- $\tau \to \infty$: $p_X \to 0.5$, which is exactly your original Randomized heuristic.
- $\tau \to 0$: $p_X \to 1$ if $\sigma_Y(v) > \sigma_X(v)$, else $p_X \to 0$. This is the greedy placement rule applied in a random vertex order.

What to run and report:

- For at least 6 benchmark graphs, run this construction (without local search) for $\tau = 0.5, 1, 5, 50$, averaging each over at least 10 runs.
- Also report the average of your original Randomized heuristic (10 runs) and the Greedy heuristic value.